

RAG-Based Customer Support Assistant – Low Level Design

1. Module Design

- PDF Ingestion – Load and parse documents
- Chunking – Split text into smaller parts
- Embedding – Convert text to vectors
- Vector Store – Store embeddings (ChromaDB)
- Retrieval – Fetch relevant chunks
- Query Processing – Handle user input
- LangGraph – Control workflow
- HITL – Enable human intervention

2. Data Structures

Document: { filename, content }

Chunk: { text, metadata, chunk_id }

Embedding: { vector, chunk_id }

Query: { user_query, response }

3. Workflow

Input → Processing → Output

4. Routing Logic

- Strong match → Answer
- Unclear query → Clarify
- No data → Fallback
- Complex → Escalate

5. HITL

- Trigger on low confidence
- Human review → Return response

6. API

Input: User query

Output: Generated response

7. Error Handling

- Empty PDF
- No matching chunks
- LLM failure